

Alessandro Santini

PHD CANDIDATE IN GRAVITATIONAL PHYSICS

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Education

Ph.D. in Physics

MAX PLANCK INSTITUTE FOR GRAVITATIONAL PHYSICS

Potsdam, Germany

2023–present

- **Supervisors:** Dr. Jonathan Gair, Prof. Dr. Alessandra Buonanno

Master's Degree in Astrophysics and Space Physics

UNIVERSITÀ DEGLI STUDI DI MILANO-BICOCCA

Milan (MI), Italy

2021–2023

- **final degree grade:** 110/110 with distinction (cum laude)
- **Thesis:** Black-hole mergers in disk-like environments could explain the observed $q - \chi_{\text{eff}}$ correlation
- **Thesis advisors:** Prof. Davide Gerosa, Dr. Roberto Cotesta

Bachelor's Degree in Physics

UNIVERSITÀ DEGLI STUDI DI MILANO-BICOCCA

Milan (MI), Italy

2018–2021

- **final degree grade:** 110/110 with distinction (cum laude)
- **Thesis:** Resolution of the Euler equations using the Athena++ code
- **Thesis advisor:** Prof. Bruno Giacomazzo

High School Diploma

LICEO SCIENTIFICO STATALE FEDERIGO ENRIQUES

Lissone (MB), Italy

2013–2018

- **Final degree grade:** 97/100

INTERNSHIPS

Research visit

OBSERVATOIRE DE LA CÔTE D'AZUR, 1 MONTH

Nice, France

May 2026

Erasmus+ Scholarship

JOHNS HOPKINS UNIVERSITY

Baltimore (MD), USA

April–July 2023

Skills

- Programming** Python (proficient) – GPU computing (CUDA, Torch, Cupy, Numba, JAX) – Bash – Mathematica (basic) – C, C++ (basic)
- Other tools** Latex (proficient), Git, Microsoft Office suite
- Languages** Italian (Native), English (Fluent)

Talks, conferences & workshops

INVITED TALKS

LISA AI Hackathon

[ESTEC, Noordwijk, The Netherlands](#)

Machine Learning in the Global Fit – DISCUSSION SESSION

July 2026

Observatoire de la Côte d’Azur

[OCA, Nice, France](#)

Advancing LISA Global fit prototypes

May 2026

ETH Zurich – Gravitational Physics Group

[ETH, Zurich, Switzerland](#)

Advancing LISA Global fit prototypes

March 2026

LISA without frontiers

[Sexten, Italy](#)

Modelling noise and characterising stochastic signals in LISA – PANEL SESSION

January 2026

CONTRIBUTED TALKS

EMRI Search and Inference within the LISA Global Fit - Part II

[NUS, Singapore](#)

Multi-source analysis of EMRIs in the Mojito light common dataset

July 2026

26th SIGRAV Meeting

[Unimib, Milan, Italy](#)

A flexible, GPU-accelerated approach for the joint characterization of LISA instrumental noise and Stochastic Gravitational Wave Backgrounds

September 2025

EMRI Search and Inference within the LISA Global Fit - Part I

[APC, Paris, France](#)

Don't reinvent the wheel: including Extreme Mass Ratio Inspirals in the LISA global fit

June 2025

Amaldi15

[Online](#)

Black-hole mergers in disk-like environments could explain the observed $q - \chi_{\text{eff}}$ correlation

July 2023

APS April Meeting

[Minneapolis, USA](#)

Migration traps in AGN disks and hierarchical mergers as promising origin of the observed $q - \chi_{\text{eff}}$ correlation

April 2023

CONTRIBUTED POSTERS

15th LISA Symposium

[UCD, Dublin, Ireland](#)

A flexible approach for the joint characterization of LISA instrumental noise and Stochastic Gravitational-Wave Backgrounds

July 2024

16th LISA Symposium

[Baltimore \(MD\), USA](#)

Advancing LISA Global Fit prototypes in the time-frequency domain

June 2026

Publication record

1. Speri, L.; Duque, F.; Barsanti, S.; **Santini, A.**; Kejriwal, S.; Burke, O.; Chapman-Bird, C. E. A.
"Quantifying the Scientific Potential of Intermediate and Extreme Mass Ratio Inspirals with the Laser Interferometer Space Antenna",
ArXiv preprint
2. De Santi, F.; **Santini, A.**; Toubiana, A.; Karnesis, N.; Gerosa, D.
"Inferring the population properties of galactic binaries from LISA's stochastic foreground", Phys.Rev.D 113, 123006
3. **Santini, A.**; Muratore, M.; Gair, J.; Hartwig, O.
"A flexible, GPU-accelerated approach for the joint characterization of LISA instrumental noise and Stochastic Gravitational Wave Backgrounds", Phys.Rev.D 112, 084050
4. Chapman-Bird, C. E. A. et al (including **Santini, A.**)
"The Fast and the Frame-Dragging: Efficient waveforms for asymmetric-mass eccentric equatorial inspirals into rapidly-spinning black holes", Phys.Rev.D 112, 104023
5. Fabbri, C. M.; Gerosa, D.; **Santini, A.**; Mould, M.; Toubiana, A.; Gair, J.
"Reconstructing parametric gravitational-wave population fits from non-parametric results without refitting the data", Phys.Rev.D 111, 104053
6. Khalvati, H.; **Santini, A.**; Duque, F.; Speri, L.; Gair, J.; Yang, H.; Brito, R.
"Impact of relativistic waveforms in LISA's science objectives with extreme-mass-ratio inspirals", Phys.Rev.D 111, 082010
7. **Santini, A.**; Gerosa, D.; Cotesta, R.; Berti, E.
"Black-hole mergers in disk-like environments could explain the observed $q - \chi_{\text{eff}}$ correlation", Phys.Rev.D 108, 083033

Public Outreach

Berlin Science Week

Berlin, Germany

November 2024

Language: English

Deciphering black hole symphonies: The new world of gravitational wave astronomy.

Speaker at one of the highlight, invited events of Berlin Science Week 2024 (webpage [here](#)), presenting the current status of gravitational-wave astronomy together with 7 other researchers from the Max Planck Institute for Gravitational Physics. Audience: > 200 people.

Long Night of Astronomy

Berlin, Germany

September 2025

Language: English

Sounds of silence: 10 Years of gravitational-wave astronomy.

Speaker at the closing event of the Long Night of Astronomy 2025 (webpage [here](#)), presenting the last 10 years of gravitational-wave astronomy together with 8 other researchers from the Max Planck Institute for Gravitational Physics. Audience: > 100 people.